

USAID AHMED

AI ENGINEER

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PROFESSIONAL SUMMARY

AI Engineer with hands-on experience building intelligent, production-ready systems across the full AI stack. Currently developing RAG pipelines, agentic AI architectures, and full-stack AI applications as an AI Intern at Saylani. Completed a one-year AI & Data Science program at Saylani Mass IT Training Program and a Data Science internship at 10Pearls, contributing to a real-world AQI forecasting pipeline. Proficient in machine learning, deep learning (ANN, CNN, GANs), NLP, Generative AI, LangChain, and LangGraph, with experience in React.js, Next.js, FastAPI, and Node.js. Pursuing a B.S. in Computer Science at the University of Karachi, focused on building scalable AI systems that solve real problems.

TECHNICAL SKILLS

AI & Machine Learning: Supervised & Unsupervised Learning, Feature Engineering, Model Training & Evaluation, Ridge Regression, Random Forest, Gradient Boosting, scikit-learn

Deep Learning: ANN, CNN, KNN, RNN, Transfer Learning, GANs (Generative Adversarial Networks), TensorFlow, Keras, PyTorch

NLP & Generative AI: Natural Language Processing, Sentiment Analysis, Text Classification, Transformers, Large Language Models (LLMs), Prompt Engineering, RAG (Retrieval-Augmented Generation), LangChain, LangGraph.

Data Science: Pandas, NumPy, Matplotlib, Seaborn, EDA, Data Cleaning & Wrangling, Web Scraping (BeautifulSoup, Requests), Parquet, Hopworks, Streamlit, Chainlit

Frontend: HTML5, CSS3, Bootstrap, Tailwind CSS, JavaScript, React.js, Next.js, React Native, Streamlit, Chainlit

Backend & APIs: Python, FastAPI, Node.js (Express), Java, C++, Redis

Databases & Vector Stores: MongoDB, Firebase, Supabase, PostgreSQL (pgAdmin), ChromaDB, Vector Databases (FAISS, Pinecone concepts)

MLOps & DevOps: GitHub Actions, CI/CD Pipelines, Dockers, REST APIs, Feature Stores

Soft Skills: Problem Solving, Team Collaboration, Adaptability, Time Management, Technical Communication, Fast Learning

EXPERIENCE

AI Engineering Intern | [Saylani](#) May 2026 – Present

- Building a full-stack AI web application inspired by Google Teachable Machine, enabling users to train custom image classification models directly in the browser.
- Developing a RAG (Retrieval-Augmented Generation) powered chatbot for FIFA, utilizing ChromaDB as a vector store and LangChain for retrieval orchestration.
- Implementing end-to-end pipelines combining FastAPI backends, React.js frontends, and LLM integrations for production-ready AI applications.
- Collaborating on multi-project AI initiatives involving LangGraph-based agentic workflows and document Q&A systems.

Tools & Tech: Python, FastAPI, LangChain, LangGraph, ChromaDB, React.js, Next.js, OpenAI API, HuggingFace, TensorFlow, Node.js, MongoDB

Data Science Intern | [10Pearls](#) 2025

- Contributed to an AQI forecasting pipeline predicting PM2.5 levels for Karachi across 24, 48, and 72-hour horizons using Open-Meteo and OpenAQ APIs.
- Built reliable data ingestion and cleaning workflows, storing processed data in Parquet format with aligned timestamps for downstream ML workflows.
- Applied feature engineering techniques and assisted in model training and evaluation using Ridge Regression, Random Forest, and Gradient Boosting.

- Developed a Streamlit dashboard to visualize predictions and model insights for stakeholder review.
- Gained hands-on exposure to CI/CD automation with GitHub Actions and feature-store integration via Hopsworks.

Tools & Tech: Python, Pandas, NumPy, scikit-learn, Streamlit, Open-Meteo API, OpenAQ API, Parquet, GitHub Actions, Hopsworks

KEY PROJECTS

Teachable Machine AI Platform

- Full-stack web app allowing users to upload image datasets, train classification models in the browser, and test predictions in real time.
- Stack: React.js, FastAPI, TensorFlow.js, Python, MongoDB.

FIFA RAG Chatbot

- Conversational AI system built with RAG architecture for answering FIFA-related questions using grounded retrieval from a curated knowledge base.
- Stack: Python, LangChain, ChromaDB, FastAPI, OpenAI API, React.js.

AQI Forecasting Pipeline (10Pearls)

- End-to-end ML pipeline for air quality index prediction with real-time API ingestion, feature engineering, and a Streamlit dashboard for visualization.

EDUCATION

B.S. Computer Science *January 2024 – Present*

University of Karachi (UBIT)

Intermediate – Pre-Engineering *September 2021 – June 2023*

Adamjee Govt. Science College, Karachi

CERTIFICATIONS & TRAINING

AI & Data Science Program *March 2025 – March 2026*

Saylani Mass IT Training Program | Completed

MERN Stack Development *November 2023 – April 2024*

Expertizo Institute

LANGUAGES

English: Professional Proficiency | Urdu: Native